



BANGLADESH UNIVERSITY OF BUSINESS & TECHNOLOGY

COMPUTER SCIENCE AND ENGINEERING

Real-Time Health Monitoring and Emergency Alert System

INTAKE-50
SECTION-04

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1. Introduction

The Real-Time Health Monitoring and Emergency Alert System is a smart IoT-based device designed to monitor vital signs such as temperature, heart rate, SpO₂, and sudden body movement (fall detection). The system provides instant alerts through LED indicators, a buzzer alarm, Blynk notifications, and email warnings. It ensures timely emergency response, especially for elderly individuals, cardiac patients, and people living alone.

2. Literature Review

Continuous health monitoring using IoT has become essential in modern healthcare. Previous research highlights the importance of remote patient observation, early disease detection, and automated emergency alerts.

Key Background Concepts:

- **IoT-Based Healthcare:** Enables remote real-time monitoring via internet connectivity.
- **Biomedical Sensors:** Sensors like MAX30100 allow low-cost physiological measurement.
- **Fall Detection Systems:** Accelerometer-based threshold detection is proven effective for accidental fall detection.
- **Cloud Platforms (Blynk):** Provides real-time dashboards and alerting systems.

Required Components:

- **ESP32** – Central controller with WiFi.
- **MAX30100** – Measures Heart Rate and SpO₂.
- **DHT11** – Measures temperature.
- **MPU6050** – Detects sudden movements and falls.
- **OLED Display** – Shows live readings.
- **Buzzer** – Emergency sound alert.
- **LEDs (3 units)** – Separate indicators for temperature, HR/SpO₂, and fall detection.
- Breadboard, jumper wires, and power supply.

3. Methodology

The system follows a step-by-step data acquisition and alert mechanism.

Methodology Steps:

1. **Sensor Initialization:** ESP32 initializes MAX30100, DHT11, and MPU6050.
2. **Continuous Monitoring:** Sensors read temperature, pulse rate, SpO₂, and fall detection.
3. **Threshold Evaluation:** ESP32 checks if the readings cross danger limits.
4. **Alert Activation:**
 - LED 1 + Buzzer → Temperature abnormal.
 - LED 2 + Buzzer → Heart rate/SpO₂ abnormal.
 - LED 3 + Buzzer → Fall detected.
5. **IoT Notification:**
 - Blynk sends push notifications.
 - Email alerts are automatically triggered.
6. **OLED Display:** Continuously shows all vital values.

Circuit Diagram

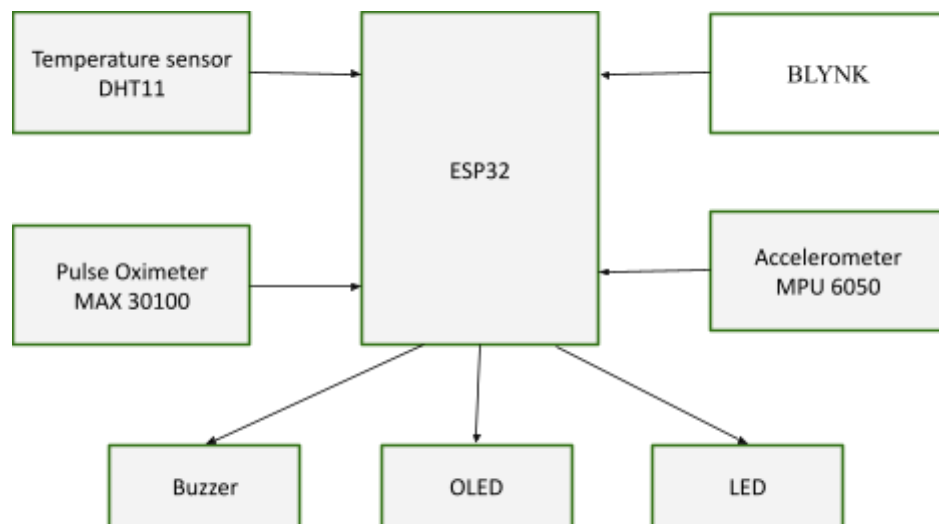


Figure: Real-Time Health Monitoring and Emergency Alert System

4. Results

The system successfully performed the following:

- Accurate measurement of **temperature, heart rate, SpO₂**.
- **Fall detection** triggered LED + buzzer instantly.
- Blynk provided **real-time notifications** and **email alerts** during critical conditions.
- OLED displayed all readings clearly.
- All three alert indicators worked as expected.

5. Conclusion

This IoT-based Real-Time Health Monitoring and Emergency Alert System is a reliable, low-cost solution for continuous health tracking. The combination of sensors, ESP32, cloud connectivity, and alert mechanisms ensures immediate awareness of emergency situations. This system can play a vital role in personal healthcare, elderly monitoring, and emergency response applications.